

$$h[n] = \delta[n] - \delta[n-4]$$

$$y[n] = x[n] * h[n]$$

$$x[n] = \cos \omega_0 n T_s$$

$$x[n] * h[n] = x[n]$$

$$y[n] = x[n] - x[n-4]$$

$$y[n] = \cos \omega_0 n T_s - \cos(\omega_0 T_s (n-4))$$

$$\cos \alpha - \cos \beta =$$

$$= -2 \sin\left(\frac{\omega_0 n T_s + \omega_0 T_s (n-4)}{2}\right) \sin\left(\frac{\omega_0 n T_s - \omega_0 T_s (n-4)}{2}\right)$$

$$\left\{ \cos \alpha - \cos \beta = -2 \sin\left(\frac{\alpha + \beta}{2}\right) \sin\left(\frac{\alpha - \beta}{2}\right) \right.$$

$$= -2 \sin \omega_0 T_s n$$

$$= -2 \sin\left(\frac{2 \omega_0 n T_s - \omega_0 T_s \cdot 4}{2}\right) \sin\left(\frac{4 \omega_0 T_s}{2}\right)$$

$$= -2 \sin(2 \omega_0 T_s) \sin(\omega_0 n T_s - 2 \omega_0 T_s)$$

$$y[n] = \underbrace{-2 \sin(2 \omega_0 T_s)}_A \cos\left(\underbrace{\omega_0 n T_s - 2 \omega_0 T_s}_{\phi} + \frac{\pi}{2}\right)$$

$$x[n] = \left(\frac{1}{2}\right)^n \cdot u[n]$$

$$y_2[n] = \left(\frac{1}{2}\right)^n \cdot u[n] - \left(\frac{1}{2}\right)^{n-4} \cdot u[n-4]$$

$$x[n] - x[n-4]$$

$$n < 0$$

$$y_2[n] = 0$$

$$4 > n \geq 0$$

$$y_2[n] = \left(\frac{1}{2}\right)^n$$

$$n \geq 4$$

$$y_2[n] = \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^{n-4} = \left(\frac{1}{2}\right)^n - \left(\frac{1}{2}\right)^n \cdot \left(\frac{1}{2}\right)^{-4} = \frac{1}{2} - \frac{1}{2} \cdot 16 = -15 \cdot \frac{1}{2}^n$$

$$y_2[n] = \left(\frac{1}{2}\right)^n \cdot \left(1 - \left(\frac{1}{2}\right)^{-4}\right) = \left(\frac{1}{2}\right)^n \cdot (-15)$$

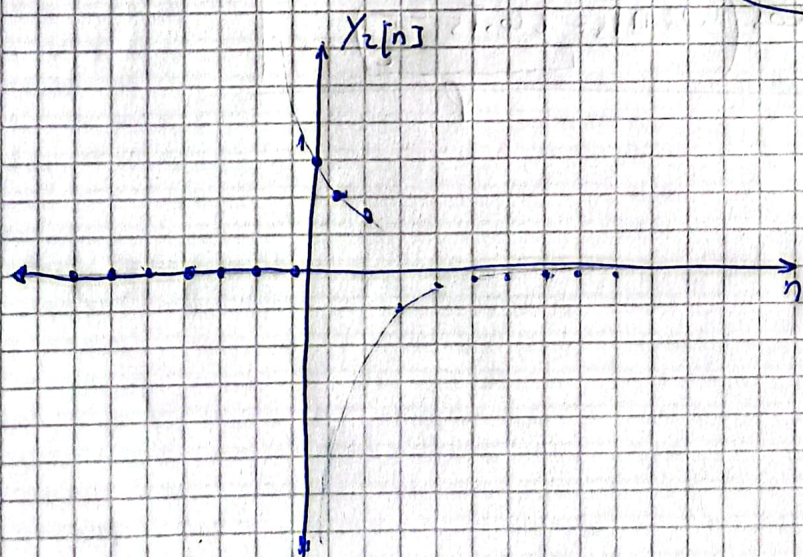
$$y_2[n] = (-15) \left(\frac{1}{2}\right)^n$$

$$u[n] = \begin{cases} 0 & n < 0 \\ 1 & n \geq 0 \end{cases}$$

$$u[n-4] = \begin{cases} 0 & n < 4 \\ 1 & n \geq 4 \end{cases}$$



$$y_2[n] = \begin{cases} 0 & n < 0 \\ \left(\frac{1}{2}\right)^n & 0 \leq n < 4 \\ (-15) \left(\frac{1}{2}\right)^n & n \geq 4 \end{cases}$$



$$X[n] = u[n+1] - u[n-2]$$

$$u[n+1] = \begin{cases} 0 & n < -1 \\ 1 & n \geq -1 \end{cases}$$

$$Y_3[n] = u[n+1] - u[n-2] - (u[n-4+1] - u[n-4-2])$$

$$u[n-2] = \begin{cases} 0 & n < 2 \\ 1 & n \geq 2 \end{cases}$$

$$Y_3[n] = u[n+1] - u[n-2] - u[n-3] + u[n-6]$$

$$\boxed{\begin{aligned} n < -1 \\ Y_3[n] &= 0 \end{aligned}}$$

$$2 > n \geq -1$$

$$Y_3[n] = 1 = u[n+1]$$

$$3 > n \geq 2$$

$$Y_3[n] = 0$$

$$6 > n \geq 3$$

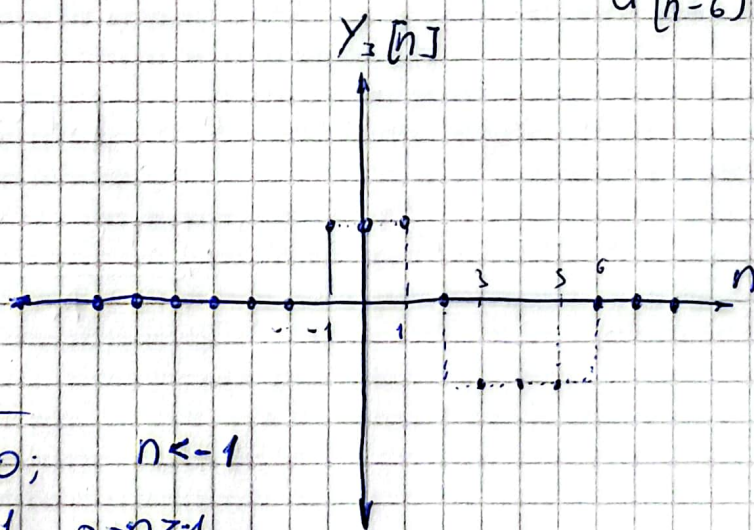
$$Y_3[n] = -1$$

$$n \geq 6$$

$$Y_3[n] = 0$$

$$u[n-3] = \begin{cases} 0 & n < 3 \\ 1 & n \geq 3 \end{cases}$$

$$u[n-6] = \begin{cases} 0 & n < 6 \\ 1 & n \geq 6 \end{cases}$$



$$Y[n] = \begin{cases} 0; & n < -1 \\ 1, & 2 > n \geq -1 \\ 0; & 3 > n \geq 2 \\ -1 & 6 > n \geq 3 \\ 0 & n \geq 6 \end{cases}$$